

Fetching Data From an API

Module 17 Study Notes • Automated Data Extraction

1. What is an API?

An **API (Application Programming Interface)** acts as a bridge or data pipeline that allows two completely different software systems to communicate. Instead of a company sharing their entire raw database (which is insecure), they write an API layer. You send a specific request to this layer, and it returns exactly the data you asked for.

Example: When you use MakeMyTrip to book a train, MakeMyTrip does not own the railway database. It uses the IRCTC API to check seat availability in real-time.

2. The Data Extraction Workflow

To pull data from an API to build a Machine Learning dataset, you follow a standard process:

A. Authentication (The API Key)

Most public APIs (like TMDB - The Movie Database) require you to create a developer account and generate an **API Key**. This unique string of characters authenticates your request and tracks your usage limits.

B. Sending the Request

In Python, you use the `requests` library to query the API endpoint URL (which includes your API key).

```
response = requests.get('https://api.themoviedb.org/3/movie/top_rated?api_key=YOUR_KEY')
```

C. Parsing the JSON Response

APIs typically return data in JSON format. You must convert this raw response into a readable Python dictionary using the `.json()` method.

```
data = response.json()
```

3. Building a Full Dataset (Pagination)

APIs rarely return thousands of records in a single request; they divide the data into "pages" (e.g., 20 movies per page). To build a complete dataset, you must automate the extraction process:

- Create an empty Pandas DataFrame.
- Write a `for` loop that iterates through all the available pages (e.g., pages 1 to 428).
- In each loop iteration: dynamically change the URL page number, fetch the JSON data, extract the specific columns you need (like ID, Title, Rating), convert that page's data into a temporary DataFrame, and append it to your main DataFrame.

4. Exporting and Portfolio Building

Once your loop finishes, you will have a massive DataFrame containing thousands of rows of real-world data.

- **Export to CSV:** Save your hard work using `df.to_csv('movies_dataset.csv')`.
- **Boost Your Profile:** Uploading custom, high-quality datasets built via APIs to platforms like **Kaggle** is an excellent way to build your Data Science portfolio and attract recruiters.